

RW4-1.0 Interface Control Document

Large Reaction Wheel - 1.0 Nms

Hardware Revision 4

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Approved By:	Systems Engineering

1. Document History

This section records all revisions made to this document.

Version	Date	Author	Description
1.0	2016-03-15	J. Hartmann	Initial release for RW4-1.0 qualification
1.1	2017-01-22	J. Hartmann	Added redundant RS-485 communication interface
1.3	2019-09-05	M. Chen	Added momentum control mode
1.5	2022-02-28	R. Okafor	Refined calibration, added jitter characterization
1.7	2025-04-18	S. Petrov	Updated to 37-pin connector, added debug channels

Distribution List

Recipient	Organization	Copy Type
Systems Engineering	AnyCompany	Controlled
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Applicable Documents

ID	Title	Rev
AC-DS-00059	RW4-1.0 Product Data Sheet	D
AC-UM-00059	RW4-1.0 User Manual	C
NASA GSFC-STD-7000A	General Environmental Verification Standard (GEVS)	-
EIA/TIA-422-B	Balanced Voltage Digital Interface Circuits	-
EIA/TIA-485-A	Balanced Multipoint Digital Interface	-
ECSS-Q-ST-70-02C	Thermal Vacuum Outgassing Test	-

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3. Scope and Applicability

This Interface Control Document (ICD) defines the mechanical, electrical, environmental, and software interfaces for the AnyCompany RW4-1.0 reaction wheel assembly, hardware revision 4. It establishes the requirements and constraints for proper integration and operation.

The RW4-1.0 is a large single-axis reaction wheel providing 1.0 Nms angular momentum storage and 100 mNm peak torque. It is designed for ADCS on medium to large spacecraft platforms. The electronics design is common with the RW4-0.4 model, with a larger flywheel for increased momentum capacity.

Flight Heritage: 144 units across 2 generations; 670+ cumulative years since 2016. The RW4-1.0 is one of the most mature products in the AnyCompany reaction wheel line.

Lifecycle Status: Active, full production. Estimated 10+ years to end-of-life. Typical lead time: 12 weeks (low risk). Demand trend: stable.

4. Mechanical Interface

4.1 Mounting Interface

The RW4-1.0 features a cylindrical housing with an integrated mounting flange. Eight (8) M5 threaded inserts are provided on a 135 mm bolt circle diameter. A minimum standoff height of 5 mm is required beneath the flange for rotor clearance. Two (2) alignment pin holes (diameter 3.0 mm H7) are at 0 and 180 degrees for rotational alignment.

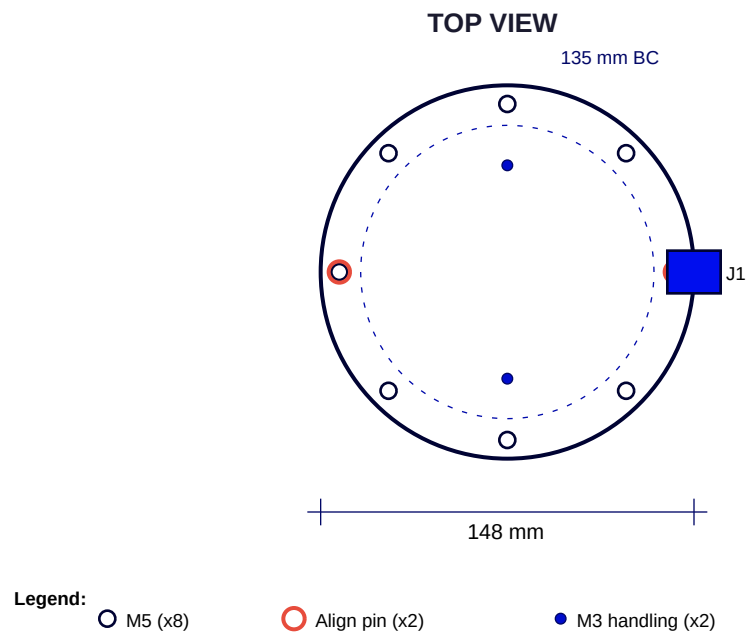


Figure 1: RW4-1.0 Mechanical Outline (Top View)

4.2 Mass and Dimensions



Figure 2: RW4-1.0 Product Photo

Parameter	Value	Units	Notes
Mass	< 1400	g	Including mounting hardware

Parameter	Value	Units	Notes
Dimensions (D x D x H)	148 x 146 x 45	mm	Cylindrical housing
Mounting	8x M5	-	135 mm bolt circle
Alignment	2x pins, 3.0 mm H7	-	0 and 180 deg
Handling	2x M3 threaded	-	For lifting fixture
Moment of Inertia	~2.1e-3	kg*m2	About spin axis

Table 1: Physical Specifications

4.3 Alignment and Handling

Two alignment pins on the mounting flange define the spin axis orientation. The pins have H7 tolerance for precise angular alignment. Two M3 threaded holes are at 90 and 270 degrees for a handling fixture.

Remove Before Flight: Kapton bearing covers (2x), handling fixture (if installed), connector cap.

5. Environmental Requirements

5.1 Thermal

Parameter	Min	Typ	Max	Units	Notes
Survival Temp	-45	-	+90	C	Unpowered
Operating Temp	-30	+20	+70	C	At mounting interface
Turn-on Temp	-30	-	+70	C	Cold start capable
Storage Temp	-50	+20	+95	C	With desiccant
Thermal Gradient	-	-	5	C/min	During operation

Table 2: Thermal Specifications

Operating temperature is defined at the mounting interface baseplate.

5.2 Pressure and Outgassing

The wheel operates in sea-level atmosphere and hard vacuum. Not qualified for intermediate pressures during ascent. Materials meet TML < 1%, CVCM < 0.1% per ECSS-Q-ST-70-02C.

5.3 Vibration

Frequency Range	Qualification	Acceptance	Duration
20 - 50 Hz	+3 dB/oct ramp	+3 dB/oct ramp	-
50 - 800 Hz	0.08 g2/Hz	0.04 g2/Hz	-
800 - 2000 Hz	-3 dB/oct rolloff	-3 dB/oct rolloff	-
Overall	20.0 Grms	14.1 Grms	3 min/axis (Q), 1 min/axis (A)

Table 3: Random Vibration Levels

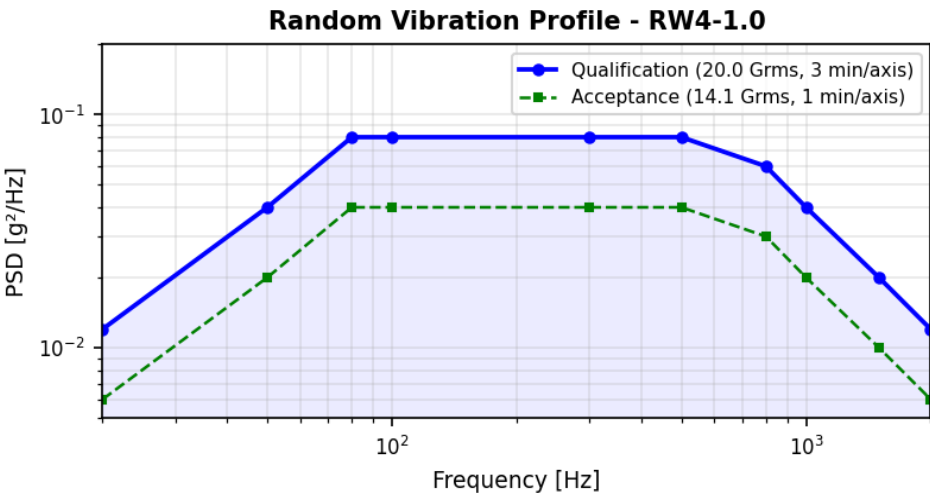


Figure 3: Random Vibration PSD Profile

5.4 Shock

Qualified to 800 G half-sine pulse, 0.5 ms duration, covering typical launch events.

5.5 Radiation

Electronics rated to 50 krad TID at board level. SEE mitigated via TMR and EDAC.

6. Electrical Interface

6.1 Power Architecture

The RW4-1.0 accepts 28 V nominal (22 to 34 V range). Internal architecture: high-efficiency DC/DC converter with EMI filter supplies 5V and 3.3V rails for electronics and a three-phase BLDC motor driver with field-oriented control.

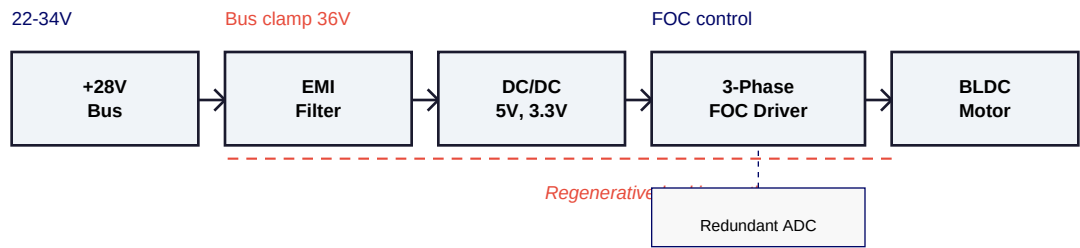


Figure 4: Power Architecture Block Diagram

Regenerative braking returns energy to the bus. Bus clamp limits to 36V.

6.3 Power Specifications

Parameter	Min	Typ	Max	Units	Notes
Input Voltage	22	28	34	V	Operating range
Abs Max Voltage	-	-	36	V	Bus clamp
Idle Power	-	5	-	W	Electronics only
Peak Power	-	-	55	W	At 100 mNm, max speed
Inrush Current	-	-	8	A	25 ms duration
Bus Capacitance	-	-	470	uF	Front-end included

Table 4: Power Specifications

6.2 Connector Pinout (37-pin Micro-D)

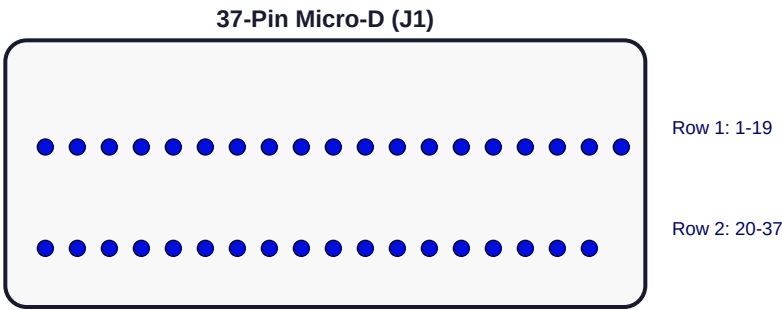


Figure 5: 37-Pin Micro-D Connector Face (J1)

Pin(s)	Signal	Color	Description	Level
1	+28V_PRI	Red	Primary power	22-34V
2	+28V_RED	Red/Wht	Redundant power	22-34V
3	RTN_PRI	Black	Primary return	GND
4	RTN_RED	Blk/Wht	Redundant return	GND
5	CHASSIS	Green	Chassis ground	GND
6	TxA+	Orange	RS-422 Pri Tx+	Diff
7	TxA-	Brown	RS-422 Pri Tx-	Diff
8	RxA+	Blue	RS-422 Pri Rx+	Diff
9	RxA-	White	RS-422 Pri Rx-	Diff
10	TxB+	Violet	RS-485 Red Tx+	Diff
11	TxB-	Gray	RS-485 Red Tx-	Diff
12	RxB+	Pink	RS-485 Red Rx+	Diff
13	RxB-	Tan	RS-485 Red Rx-	Diff
14	INHIBIT_A	Yellow	Active low inhibit A	0/28V
15	INHIBIT_B	Yel/Wht	Active low inhibit B	0/28V
16	TELEM_SYNC	Cyan	Telem sync pulse	3.3V
17	ANALOG_SPD	Wht/Org	Speed analog output	0-5V
18	ANALOG_CUR	Wht/Blu	Current analog output	0-5V
19	ANALOG_TMP	Wht/Grn	Temp analog output	0-5V
20-21	HEATER A/B	Red/Wht	Heater enables	0/28V
22	DEBUG_TX	Blu/Wht	Debug UART Tx	3.3V
23	DEBUG_RX	Grn/Wht	Debug UART Rx	3.3V
24-37	SPARE	-	Reserved	-

Table 5: 37-Pin Connector Pinout (J1)

6.4 Signal Interface

Parameter	Value	Notes
RS-422 Baud	460800	EIA/TIA-422-B
RS-422 Term.	120 ohm	Internal
Max Cable	15 m	Shielded twisted pair

Parameter	Value	Notes
RS-485 Baud	460800	Half-duplex, redundant
Debug UART	115200/8N1	Ground testing only
Analog Out	0-5V, 12-bit	Speed/current/temp

Table 6: Signal Specifications

7. Software Interface

7.1 Protocol Structure

The RW4-1.0 uses a custom binary packet protocol over RS-422 (primary) or RS-485 (redundant).



Max: 1030B | Little-endian | CRC32 (IEEE 802.3)

Figure 6: Binary Protocol Frame Structure

7.2 Message Identifiers

Code	Name	Dir	Description	Payload
0x01	HEARTBEAT	Reply	Status (auto 1 Hz)	12 B
0x02	SET_MODE	Cmd	Set operating mode	1 B
0x03	SET_SPEED	Cmd	Speed setpoint (RPM)	4 B
0x04	SET_TORQUE	Cmd	Torque setpoint (mNm)	4 B
0x05	SET_MOMENTUM	Cmd	Momentum setpoint (Nms)	4 B
0x11	TELEM_HK	Reply	Housekeeping (10 Hz)	70 B
0x12	TELEM_SCIENCE	Reply	Science (200 Hz)	1060 B
0x20	CONFIG_READ	Cmd/Reply	Read config param	4-8 B
0x21	CONFIG_WRITE	Cmd	Write config param	8 B
0x22	CONFIG_STORE	Cmd	Store to NVM	0 B
0x30	FW_UPDATE_START	Cmd	Begin FW update	4 B
0x31	FW_UPDATE_DATA	Cmd	FW data block	256 B
0x32	FW_UPDATE_VERIFY	Cmd	Verify FW CRC	4 B
0x40	SELF_TEST	Cmd/Reply	Built-in self test	0/32 B
0x41	CALIBRATE	Cmd/Reply	Calibration routine	0/16 B
0x42	DESAT_SEQUENCE	Cmd	Desaturation helper	8 B
0x43	ZERO_MOMENTUM	Cmd	Zero momentum cmd	0 B
0xF0	ERROR	Reply	Error report	4 B
0xFF	RESET	Cmd	Software reset	0 B

Table 7: Message Identifiers

7.3 Operating Modes

Mode	Value	Description	Entry
STANDBY	0x00	Motor unpowered, telem active	Power-up default
SPEED_CTRL	0x01	Closed-loop speed control	SET_MODE
TORQUE_CTRL	0x02	Open-loop torque command	SET_MODE
MOMENTUM_CTRL	0x03	Closed-loop momentum	SET_MODE
SAFE	0x04	Fault recovery, motor braked	Fault or cmd
FW_UPDATE	0x05	Firmware update	FW_UPDATE_START
SELF_TEST	0x06	Built-in test	SELF_TEST cmd

Table 8: Operating Modes

7.4 Sign Convention

Positive rotation: CCW viewed from mounting face (connector side). Positive torque accelerates in the positive direction. Positive momentum = positive spin direction.

8. Telemetry Definitions

8.1 Housekeeping Telemetry (10 Hz)

HK packet (MSG_ID 0x11) transmitted at 10 Hz when active.

Offset	Field	Type	Units	Range
0	wheel_speed_rpm	float32	RPM	+/- 4500
4	torque_actual_mNm	float32	mNm	+/- 100
8	current_phaseA_mA	float32	mA	0-2000
12	current_phaseB_mA	float32	mA	0-2000
16	current_phaseC_mA	float32	mA	0-2000
20	bus_voltage_V	float32	V	0-40
24	bus_current_A	float32	A	0-5
28	pcb_temp_C	float32	C	-50 to +100
32	motor_temp_C	float32	C	-50 to +120
36	bearing_temp_C	float32	C	-50 to +100
40	fault_register	uint32	-	Bitfield
44	mode_state	uint8	-	See Table 8
45	uptime_sec	uint32	s	0-2 ³²
49	cumulative_revs	uint64	revs	0-2 ⁶⁴
57	power_mWh	uint32	mWh	Cumulative
61	momentum_Nms	float32	Nms	+/- 1.2
65	jitter_arcsec_rms	float32	arcsec	0-10
69	bearing_preload	uint8	-	0=nom, 1=warn

Table 9: HK Telemetry (70 bytes)

8.2 Science Telemetry (200 Hz)

Science packet (MSG_ID 0x12) at 200 Hz. Includes 512-bin FFT from onboard 1024-point accelerometer sampling for micro-vibration characterization.

Offset	Field	Type	Units	Notes
0	timestamp_us	uint64	us	Microsecond
8	speed_rpm	float64	RPM	High-precision
16	torque_mNm	float64	mNm	High-precision

Offset	Field	Type	Units	Notes
24	vibration_x_mg	float32	mg	X-axis accel
28	vibration_y_mg	float32	mg	Y-axis accel
32	vibration_z_mg	float32	mg	Z-axis accel
36	spectral_bins[512]	uint16[512]	counts	FFT magnitude

Table 10: Science Telemetry (1060 bytes)

9. Fault Register

32-bit bitfield in every HK packet. Any set bit triggers SAFE mode (unless masked).

Bit	Name	Description	Recovery
0	OC_PHASE_A	Phase A overcurrent	Auto-retry 1s
1	OC_PHASE_B	Phase B overcurrent	Auto-retry 1s
2	OC_PHASE_C	Phase C overcurrent	Auto-retry 1s
3	OV_BUS	Bus overvoltage (>36V)	Wait for recovery
4	UV_BUS	Bus undervoltage (<22V)	Wait for recovery
5	OT_PCB	PCB overtemp (>85C)	Reduce torque
6	OT_MOTOR	Motor overtemp (>100C)	Reduce speed
7	OT_BEARING	Bearing overtemp (>80C)	Reduce speed
8	HALL_FAULT	Hall sensor error	SAFE, manual reset
9	ENCODER_FAULT	Encoder error	SAFE, manual reset
10	COMM_TIMEOUT_PRI	Primary timeout (>2s)	Switch to redundant
11	COMM_TIMEOUT_RED	Redundant timeout (>5s)	SAFE mode
12	MOTOR_STALL	Stall detected	Retry x3, then SAFE
13	SPEED_RUNAWAY	Speed >110% limit	Emergency brake
14	MEMORY_FAULT	EDAC uncorrectable	Reset
15	WATCHDOG_RESET	Watchdog triggered	Auto-restart
16-31	EVENT_COUNTER	Cumulative fault count	Informational

Table 11: Fault Register Bitfield (32-bit)

10. Register / Memory Map

Accessible via CONFIG_READ (0x20) and CONFIG_WRITE (0x21). All little-endian.

Address	Field	Type	Access	Description
0x0000	speed_setpoint	float32	RW	Speed setpoint (RPM)
0x0004	torque_setpoint	float32	RW	Torque setpoint (mNm)
0x0008	momentum_setpoint	float32	RW	Momentum setpoint (Nms)
0x000C	speed_measured	float32	R	Measured speed (RPM)
0x0010	torque_measured	float32	R	Measured torque (mNm)
0x0014	momentum_measured	float32	R	Measured momentum (Nms)
0x0018	bus_voltage	float32	R	Bus voltage (V)
0x001C	bus_current	float32	R	Bus current (A)
0x0020	fault_register	uint32	R	Active faults (Table 11)
0x0024	fault_mask	uint32	RW	Fault mask (1=masked)
0x0028	mode_state	uint8	R	Current mode
0x0029	hw_revision	uint8	R	Hardware rev (4)
0x002A	fw_version	uint16	R	Firmware version
0x002C	serial_number	uint32	R	Serial number
0x0030	cumulative_revs	uint64	R	Total revolutions
0x0038	operating_hours	uint32	R	Total operating hours
0x003C	speed_kp	float32	RW	Speed loop Kp
0x0040	speed_ki	float32	RW	Speed loop Ki
0x0044	torque_limit	float32	RW	Max torque (mNm)
0x0048	speed_limit	float32	RW	Max speed (RPM)
0x004C	watchdog_timeout	uint16	RW	Watchdog (ms)
0x004E	telem_rate_div	uint8	RW	Telem rate divider

Table 12: Configuration Register Map

11. Micro-Vibration Budget

The RW4-1.0 generates micro-vibrations from rotor imbalance, bearing imperfections, and motor cogging. Primary disturbance frequencies are harmonics of the rotation frequency (1x, 2x, 3x). The onboard FFT (1024-point, 512-bin) enables real-time spectral analysis.

RPM	1x Hz	1x mN	2x Hz	2x mN	3x Hz	3x mN
1000	16.7	5.0	33.3	2.0	50.0	0.8
2000	33.3	18.0	66.7	7.2	100.0	2.9
3000	50.0	38.5	100.0	15.4	150.0	6.2
4000	66.7	65.0	133.3	26.0	200.0	10.4

Table 13: Micro-Vibration Budget

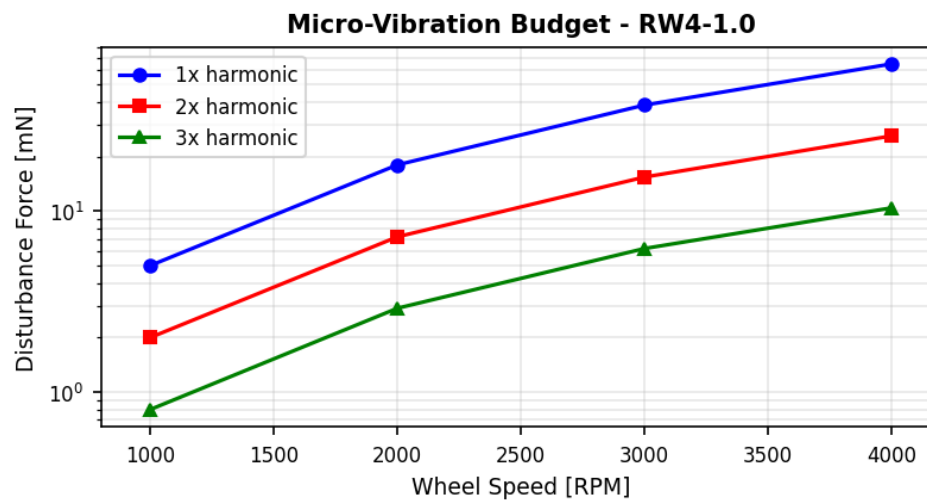


Figure 7: Micro-Vibration Forces vs Speed

Disturbance forces increase approximately with the square of speed. For jitter-sensitive payloads, operation below 2000 RPM is recommended. Science telemetry enables per-unit characterization.